
Symmetry and analysis: From Lie groups to spherical harmonics

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1 Homogeneous spaces and topological groups

Objective: To provide the basic concepts of

1. group actions
2. homogeneous spaces
3. Haar measures
4. modular homeomorphism

The concept of a homogeneous space can be found in various areas of mathematics. From differential geometry, through algebraic geometry, to areas such as harmonic analysis, the classical cases $(\mathbb{R}, S^1)$ can be treated as particular examples of harmonic analysis on homogeneous and symmetric spaces. It is exactly this perspective that we are interested in covering in these notes; although it can be taken from a very general perspective on a fairly large family of groups such as locally compact ones, we wish to focus on the classical examples of groups in which we have Lie groups.

1.1 Locally compact groups and Lie groups

We can begin the topic from the perspective of locally compact topological groups (**LCG**); it turns out that Lie groups are particular cases of this setting.

Definition 1.1.1 (Locally Compact Space). *A Hausdorff space X is said to be **locally compact** if every point $x \in X$ has a neighborhood that is compact.*

Equivalently, X is locally compact if for every $x \in X$ and every open neighborhood U of x , there exists an open set V with compact closure such that:

$$x \in V \subseteq \overline{V} \subseteq U.$$

Definition 1.1.2. *Let $(G, \cdot)$ be a group. If τ is a topology on G , it is said that G is a topological group if with the topology τ the following operations are continuous*

$$\begin{array}{ll} \cdot : G \times G \rightarrow G & i : G \rightarrow G \\ (g_1, g_2) \mapsto g_1 \cdot g_2 & g \mapsto g^{-1} \end{array}$$

Definition 1.1.3. *A **Lie group** G is a differentiable manifold G which has a group structure $(G, \cdot)$ such that the product and the inversion*

$$\begin{array}{ll} \cdot : G \times G \rightarrow G & i : G \rightarrow G \\ (g_1, g_2) \mapsto g_1 \cdot g_2 & g \mapsto g^{-1} \end{array}$$

are smooth functions.

In the previous definition, it is sufficient to require that the product be a smooth function to obtain the Lie group structure; for a partial solution of this see Problem 7-3 of [Lee12].

Example 1. $\mathbb{R}^n$ with the operation given by addition and its usual topological and differential structures is a Lie group.

Proposition 1.1.4. *The set of invertible matrices, $GL(n, \mathbb{F})$, is a Lie group.*

Proposition 1.1.5. *The orthogonal group*

$$O(n) := \{A \in GL(n, \mathbb{R}) : AA^T = A^T A = I\}$$

and the special orthogonal group

$$\mathrm{SO}(n) := \{A \in \mathrm{O}(n) : \det A = 1\}$$

are Lie groups.

Proposition 1.1.6. *Let $A \in M_n(\mathbb{R})$, and T be the linear transformation associated with that matrix. The following statements are equivalent:*

- $A \in \mathrm{O}(n)$,
- the rows and columns of A are orthonormal,
- T is an isometry with respect to the usual inner product of $\mathbb{R}^n$,

Proposition 1.1.7. *The group $\mathrm{SO}(n)$ is connected. Moreover, it is the connected component of the identity of $\mathrm{O}(n)$.*

Proof. The proof that the group $\mathrm{SO}(n)$ is connected can be found in [WA67]. The fact that it is a connected component follows from the fact that $\mathrm{SO}(n)$ is a subgroup of index 2 in $\mathrm{O}(n)$ and that the determinant function is continuous. ■

Proposition 1.1.8. *The groups $\mathrm{O}(n)$ and $\mathrm{SO}(n)$ are compact.*

Example 2. We define the affine group as

$$G = \{(x, y) \in \mathbb{R}^2 : x > 0\}$$

with the group operation given by:

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2, x_1 y_2 + y_1)$$

This group can be realized as a matrix group with the following "representation"

$$(x, y) \mapsto \begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix}$$

The inverse element is:

$$\begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1/x & -y/x \\ 0 & 1 \end{pmatrix}$$

1.2 Radon and Haar Measures

The concept of Haar measure will allow us to connect our homogeneous spaces with the theory of functional analysis and harmonic analysis.

Definition 1.2.1. *Let X be a locally compact Hausdorff space. A **Radon measure** is a Borel measure on X that is*

- finite on compact sets,
- inner regular on open sets, and
- outer regular on Borel sets.

Definition 1.2.2. *Let G be a locally compact Hausdorff group. A **left Haar measure** is a Radon measure μ such that*

$$\mu(xE) = \mu(E)$$

for every Borel set E , and every $x \in G$.

Example 3. The Lebesgue measure on $\mathbb{R}^n$ is a Haar measure for the group $(\mathbb{R}^n, +)$.

Example 4. The Dirac measure centered at the origin is not a left Haar measure for the group $(\mathbb{R}^n, +)$.

Given a topological group G , it may happen that there is no Haar measure associated with said group; however, it turns out that for locally compact groups, Haar measures always exist.

Theorem 1.2.3. A Course in Abstract Harmonic Analysis[Fol94, Theorem 2.10] *Every locally compact group G has a left Haar measure, and this measure is unique up to multiplication by a scalar.*

1.3 Homogeneous spaces and group actions.

A homogeneous space is a space which locally looks identical at any point. We are interested in knowing what a homogeneous space is, because it is on these spaces that the representations we will study are carried out. There are at least two ways to study homogeneous spaces: a geometric one (homogeneous manifolds) and another in which we only stay at a topological level. For both, it is necessary to have the concept of a group action.

Definition 1.3.1. Let G be a group (not necessarily topological) and X a set. A **left action** of G on X is a function $\varphi : G \times X \rightarrow X$, which we denote by $(g, x) \mapsto g \cdot x$, with the following properties:

- $g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x \quad (\forall x \in X) \quad \text{and} \quad (\forall g_1, g_2 \in G),$
- $1 \cdot x = x \quad (\forall x \in X).$

When we have a left action of a group G on a space X , we will say that G **acts on the left on X** and we will denote it as $G \curvearrowright X$.

In case G is a locally compact Hausdorff topological group and X is a locally compact Hausdorff topological space, we will additionally require that the function φ be continuous and that

$$\begin{aligned} \varphi_g : X &\rightarrow X \\ x &\mapsto g \cdot x \end{aligned}$$

be a homeomorphism for each $g \in G$.

We are interested in a particular type of actions in order to define a homogeneous space.

Definition 1.3.2. Let G be a group acting on the left on a space X . We will say that the action is **transitive** if for any $x, y \in X$ there exists a $g \in G$ such that

$$x = g \cdot y.$$

Note that in the previous definition we did not require the group to be a Lie group, or even a topological group. When the group G is a topological group and X is a topological space, we will require the action to be a continuous function. In the case of a Lie group G and X a smooth manifold, we will require the action to be a smooth function between manifolds.

Definition 1.3.3. Let G be a group acting on a space X and a fixed $w \in X$. We define **the isotropy group of w** as

$$G^w = \{g \in G : g \cdot w = w \quad \text{para todo } g \in G\}.$$

It is possible to prove that G^w is indeed a group; this group will be useful to us later for the main result of the work. Now we are in a position to define what a homogeneous space is.

Definition 1.3.4. A **homogeneous space (smooth)** is a smooth manifold M together with the smooth and transitive action of a Lie group G on M . A **homogeneous space** is a locally compact topological space M together with the transitive action of a locally compact topological group G , which satisfies the condition that M is homeomorphic to a quotient space G/H , with H some closed subgroup of G .

Example 5. The action of the group $O(n)$ on S^{n-1} , given by left multiplication

$$A \cdot x = Ax,$$

is transitive, since given a vector x of norm 1, we can complete it to an orthogonal basis of $\mathbb{R}^n$ and define an orthogonal matrix whose columns are the vectors of this basis, such that

$$A \cdot e_n = Ae_n = x.$$

Therefore S^{n-1} is a homogeneous space of $O(n)$.

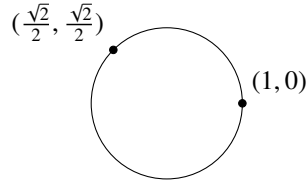
Example 6. The action of the group $SO(n)$ on S^{n-1} given by left multiplication is again transitive; therefore S^{n-1} is a homogeneous space of $SO(n)$. The proof that the action is transitive is analogous to the previous example.

Example 7. The matrix

$$g = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

is orthogonal. Let's see that for an $(x, y) \in \mathbb{R}^2$, we have

$$g \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \cos(\theta) - y \sin(\theta) \\ x \sin(\theta) + y \cos(\theta) \end{bmatrix}.$$



Example 8. The group $SL(2, \mathbb{R})$ acts smoothly on the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$, by means of Möbius transformations, that is,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot z = \frac{az + b}{cz + d}.$$

This action is transitive, since, given a complex $z = x + iy \in \mathbb{H}$, we consider the matrix

$$A = \begin{bmatrix} \sqrt{y} & \frac{x}{\sqrt{y}} \\ 0 & \frac{1}{\sqrt{y}} \end{bmatrix}$$

which satisfies $A \cdot i = z$. Therefore, the action is transitive and the upper half-plane is a homogeneous space of $SL(2, \mathbb{R})$.

The following theorem gives us a way to construct homogeneous spaces.

Theorem 1.3.5. [Lee12, Theorem 21.17] *Let G be a Lie group and H a closed subgroup of G . The quotient space G/H is a topological manifold and has a unique smooth structure such that the quotient map*

$$\begin{aligned} \pi : G &\rightarrow G/H \\ g &\mapsto gH \end{aligned}$$

is a smooth submersion. The left action given by

$$g_1 \cdot (g_2 H) = (g_1 g_2) \cdot H$$

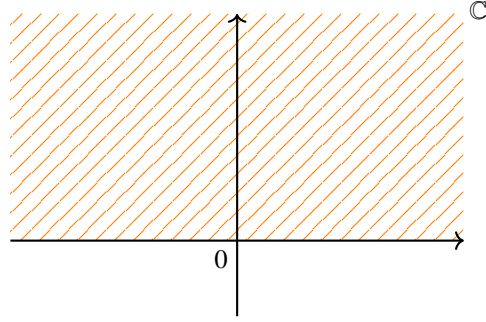


Figure 1.1: Cartesian plane with shaded upper area and diagonal lines.

makes the space G/H a homogeneous space.

The main theorem of homogeneous spaces tells us that a homogeneous space can be identified with a quotient of Lie groups.

Theorem 1.3.6. [Lee12, Theorem 21.18] Let G be a Lie group, M a homogeneous space of G , and $p \in M$. The isotropy group of G , G_p , is a closed subgroup of G and furthermore the transformation

$$\begin{aligned} F : G/G_p &\rightarrow M \\ gG_p &\mapsto g \cdot p \end{aligned}$$

is a diffeomorphism.

Note that with this theorem we can express S^{n-1} as a quotient of Lie groups.

Proposition 1.3.7. Consider S^{n-1} and $\mathrm{SO}(n) \curvearrowright S^{n-1}$ acting by left multiplication. The isotropy group of $e_n = (0, \dots, 1)$ is isomorphic to $\mathrm{SO}(n-1)$ and therefore

$$S^{n-1} \cong \mathrm{SO}(n)/\mathrm{SO}(n-1) \cong \mathrm{O}(n)/\mathrm{O}(n-1).$$

Proof. It is possible to show using Proposition 1.1.6 that

$$\mathrm{SO}(n)_{e_n} = \left\{ \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SO}(n-1) \right\} \cong \mathrm{SO}(n-1).$$

The equality $Ae_n = e_n$ implies that the last column of A must be e_n . Using the orthogonality of the rows, we have that the last row must be $e_n^T = (0, \dots, 1)$ and that the rest is an orthogonal matrix of size $n-1$. Similarly, it is shown that

$$\mathrm{O}(n)_{e_n} \cong \mathrm{O}(n-1). \blacksquare$$

1.4 Modular functions

Given a homogeneous space G/H , with G a locally compact group, we are interested in “inheriting” the Haar measure from G to the homogeneous space. The conditions for this to happen are expressed in terms of modular functions.

Definition 1.4.1 (Modular function). Let G be a locally compact group with left Haar measure λ . For each $x \in X$ we define the function

$$\lambda_x(E) = \lambda(Ex).$$

This is a new Haar measure, since $y(Ex) = (yE)x$. By the uniqueness of the Haar measure (Theorem 1.2.3), there exists a number $\Delta(x) > 0$ such that $\lambda_x = \Delta(x)\lambda$. The function

$$\begin{aligned}\Delta : G &\rightarrow \mathbb{R}^+ \\ x &\mapsto \Delta(x)\end{aligned}$$

is called the *modular function of G* .

The modular function essentially measures how far the left Haar measure is from being a right Haar measure.

Proposition 1.4.2. *If K is a compact subgroup of G then $\Delta_K \equiv 1$. Where Δ_K represents the Haar measure of the group K .*

Theorem 1.4.3. [Fol94, Theorem 2.51] *Let G be a locally compact Hausdorff group and H a closed subgroup of G . There exists a G -invariant Radon measure μ on G/H if and only if $\Delta_G|_H = \Delta_H$. Furthermore, this measure is unique up to multiplication by a scalar.*

A corollary of this theorem is the following.

Proposition 1.4.4. *If H is compact, G/H has a G -invariant Radon measure.*

Although this last result can be obtained in a much simpler way, I prefer to use this theorem as an excuse to present the modular morphism.

Note that, in particular, this implies that there are invariant measures under the action of the group $\mathrm{SO}(n+1)$ on S^n and for the action of the group $\mathrm{SL}(2, \mathbb{R})$ on the upper half-plane.

2 Reproducing Kernel Hilbert spaces

Objectives:

1. Define the concept of *RKHS*
2. Give the good reproducing properties
3. Give 2 concrete examples; from these examples arise some special functions

As we will see later, the problem of proving that a representation of a group G on a vector space V is irreducible is not simple. However, for certain types of vector spaces, there are very concrete conditions that guarantee that such a representation is irreducible. An example of these spaces are the **reproducing kernel Hilbert spaces**.

2.1 Definition and examples

Definition 2.1.1. Let X be a non-empty set. Let us denote by $\mathcal{F}(X, \mathbb{C})$ the space of functions from X to $\mathbb{C}$. We will say that a Hilbert space $\mathcal{H} \subseteq \mathcal{F}(X, \mathbb{C})$ is a **reproducing kernel Hilbert space (RKHS)** if for each $x_0 \in X$, the evaluation functional at x_0

$$\begin{aligned} F_{x_0} : \mathcal{H} &\rightarrow \mathbb{C} \\ f &\mapsto f(x_0) \end{aligned}$$

is bounded.

Suppose that $\mathcal{H} \subseteq \mathcal{F}(X, \mathbb{C})$ is an RKHS, taking $y \in X$ let us consider the evaluation functional at the point $y \in X$:

$$\begin{aligned} F_y : \mathcal{H} &\rightarrow \mathbb{C} \\ f &\mapsto f(y). \end{aligned}$$

By the Riesz representation theorem, there exists a vector $k_y \in \mathcal{H} \subseteq \mathcal{F}(X, \mathbb{C})$ such that

$$F_y(f) = \langle f, k_y \rangle \quad \text{para todo } f \in \mathcal{H}.$$

We define the **reproducing kernel** of $\mathcal{H}$ as the function

$$\begin{aligned} k : X \times X &\rightarrow \mathbb{C} \\ (x, y) &\mapsto k(x, y) = k_y(x). \end{aligned}$$

Example 9. Taking $X = \{1, 2, \dots, n\}$ we identify $\mathcal{F}(\{1, \dots, n\}, \mathbb{C})$ with $\mathbb{C}^n$. Let $\mathcal{H} \subseteq \mathbb{C}^n$ be arbitrary, $\mathcal{H}$ is a reproducing kernel Hilbert space, since every linear functional on $\mathbb{C}^n$ is bounded. In particular, for $j \in X$ the functional

$$F_j(x) = F_j((x_1, \dots, x_n)) = x_j$$

is bounded, of course $F_1, \dots, F_n$ is the dual basis of the canonical basis of $\mathbb{C}^n$. Furthermore, let us note that

$$F_j(z) = \langle z, e_j \rangle.$$

The vectors of $\mathcal{H} = \mathbb{C}^n$ are functions defined on the set $X = \{1, \dots, n\}$. We take $j \in X$ and the functional

$$F_j(z) = z_j = \langle z, e_j \rangle,$$

then $k_j = e_j$; therefore, the reproducing kernel is

$$\begin{aligned} k : X \times X &\rightarrow \mathbb{C} \\ (i, j) &\mapsto k_j(i) = e_j(i) = \delta_{ij}. \end{aligned}$$

Example 10. Let $X = \mathbb{N}$ and $d\mu$ be the counting measure. Consider the Hilbert space

$$\mathcal{H} = L^2(X, d\mu) = \ell^2(X) \subseteq \mathcal{F}(X, \mathbb{C}),$$

that is, the space of square-summable sequences; that is, the space of sequences

$$X = (x_n)_{n=1}^{\infty} \quad \text{such that} \quad \sum_{n=1}^{\infty} |x_n|^2 < \infty.$$

The inner product is given by

$$\langle x, y \rangle = \sum_{n=1}^{\infty} x_n \overline{y_n}.$$

For each $j \in \mathbb{N}$ we define the vector $e_j \in \ell^2(\mathbb{N})$ as

$$e_j(i) = \delta_{ij}.$$

In the space $\mathcal{H}$, the evaluation functional at j takes the form

$$\begin{aligned} F_j : \ell^2(\mathbb{N}) &\rightarrow \mathbb{C} \\ x &\mapsto x(j) = x_j. \end{aligned}$$

Let us see that

$$F_j(x) = x_j = \langle x, e_j \rangle.$$

By the uniqueness in the Riesz representation theorem, we have that $K_j(\cdot) = e_j(\cdot)$. Thus, the reproducing kernel is

$$\begin{aligned} K(i, j) &= \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases} \\ &= e_j(i). \end{aligned}$$

Example 11. Consider $D \subseteq \mathbb{C}$ an open and connected set distinct from $\mathbb{C}$. We have the space of holomorphic functions on D which we denote by $\mathbb{H}(D)$. Taking a measure $d\mu$ on D which can be, for example, the Lebesgue measure $dx dy$, we define the **Bergman space** as the intersection

$$\mathbb{H}(D) \cap L^2(D, d\mu).$$

The classical case is when the measure $d\mu$ is the Lebesgue measure. In this case the Bergman space is an RKHS, the reproducing kernel is called the **Bergman kernel** and is a function $K : D \times D \rightarrow \mathbb{C}$ such that

$$f(z) = \int_D f(w) K(z, w) d\mu(w) = \langle f, K_z \rangle,$$

where $K_z(w) = K(w, z)$ para todo $w \in D$.

If $d\mu$ is the Lebesgue measure and D is the unit disk, we can explicitly express the reproducing kernel as

$$K(z, w) = -\frac{1}{\pi(1 - z\overline{w})^2}.$$

Not every L^2 space is an RKHS; for this, let us consider the following counterexample.

Definition 2.1.2. A **multi-index** with n components is an element of $\mathbb{Z}_+^n$, where $\mathbb{Z}_+$ denotes the non-

negative integers. Let $\alpha = (\alpha_1, \dots, \alpha_n)$ be a multi-index. The **length** or **height** of the multi-index is defined as the number

$$|\alpha| = \alpha_1 + \dots + \alpha_n.$$

We will denote by $\alpha!$ the number $\alpha_1! \cdots \alpha_n!$

Given $x \in \mathbb{R}^n$ and a multi-index α with n components and height m , we say that

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

is the **monomial** of degree m in n variables. Finally, we define the differential operator ∂^α by means of the formula

$$\partial^\alpha = \frac{\partial^{\alpha_1}}{\partial^{\alpha_1} x_1} \cdots \frac{\partial^{\alpha_n}}{\partial^{\alpha_n} x_n}.$$

2.1.1 Some polynomial spaces

Definition 2.1.3. We denote by $\mathcal{P}_m(\mathbb{R}^n) \subset \mathbb{C}[x_1, \dots, x_n]$ the set of homogeneous polynomials of degree m , in n real variables, with complex coefficients, that is, the space of polynomials of the form

$$\sum_{|\alpha|=m} a_\alpha x^\alpha \quad a_\alpha \in \mathbb{C}.$$

We denote by $\mathcal{H}_m(\mathbb{R}^n) \subset \mathcal{P}_m(\mathbb{R}^n)$ the set of homogeneous harmonic polynomials, that is, the polynomials P such that $\Delta P = 0$, where Δ is the Laplacian operator.

Let us note the behavior of these polynomials with respect to restrictions.

Proposition 2.1.4. The restriction map

$$\begin{aligned} R : \mathcal{P}_m(\mathbb{R}^n) &\rightarrow \mathcal{P}_m(S^{n-1}) \\ p &\mapsto p|_{S^{n-1}} \end{aligned}$$

where $\mathcal{P}_m(S^{n-1}) = \{p|_{S^{n-1}} : p \in \mathcal{P}_m(\mathbb{R}^n)\}$, is injective.

Proof. Let $p, q \in \mathcal{P}_m(\mathbb{R}^n)$ be such that they coincide on all vectors of S^{n-1} , then para todo $x \neq 0$ we have that

$$p(x) = p\left(\|x\| \cdot \frac{x}{\|x\|}\right) = \|x\|^m p\left(\frac{x}{\|x\|}\right) = \|x\|^m q\left(\frac{x}{\|x\|}\right) = q(x).$$

In the case $x = 0$, the proposition is trivial. ■

Analogously we define

$$\mathcal{H}_m(S^{n-1}) = \{p|_{S^{n-1}} : p \in \mathcal{H}_m(\mathbb{R}^n)\},$$

this set is of importance in various areas of mathematics and physics.

Definition 2.1.5 (Spherical harmonics). We will call spherical harmonics of degree m the elements of the set $\mathcal{H}_m(S^{n-1})$.

Again, the restriction operator is injective, so we can identify both spaces.

Definition 2.1.6 (Quasi-regular representation.). The groups $O(n)$ and $SO(n)$ act on $\mathbb{R}^n$ and S^{n-1} by means of left multiplication. This action generates representations of these groups on the function spaces $\mathbb{C}^{\mathbb{R}^n} = \{f : \mathbb{R}^n \rightarrow \mathbb{C}\}$ and $\mathbb{C}^{S^{n-1}} = \{f : S^{n-1} \rightarrow \mathbb{C}\}$, by means of

$$\pi(g)f(x) = f(g^{-1} \cdot x).$$

We will call this representation the **quasi-regular representation** of $O(n)$ on $\mathbb{C}^{\mathbb{R}^n}$ and of $SO(n)$ on $\mathbb{C}^{S^{n-1}}$.

The polynomial spaces defined above are subspaces of $\mathbb{C}^{\mathbb{R}^n}$, and their restrictions to the sphere are subspaces of $\mathbb{C}^{S^{n-1}}$. Moreover, they are invariant subspaces of the quasi-regular representation.

Proposition 2.1.7. *The space $\mathcal{P}_m(\mathbb{R}^n)$ is an invariant subspace of the quasi-regular representation of $O(n)$ on $\mathbb{C}^{\mathbb{R}^n}$.*

Proof. By the linearity of the representation, it is sufficient to prove the invariance only for the monomials x^{α_0} , with $|\alpha_0| = m$. First, let us see that for any matrix $g \in O(n)$ we have that

$$g^{-1}x = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n a_{1i}x_i \\ \vdots \\ \sum_{i=1}^n a_{ni}x_i \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix},$$

so that

$$\pi(g)x^{\alpha_0}(x) = x^{\alpha_0}(g^{-1}x) = y_1^{\alpha_1} \cdots y_n^{\alpha_n} = \left(\sum_{i=1}^n a_{1i}x_i \right)^{\alpha_1} \cdots \left(\sum_{i=1}^n a_{ni}x_i \right)^{\alpha_n}. \quad (2.1)$$

By the multinomial formula we have that

$$\left(\sum_{j=1}^n a_{ij}x_j \right)^{\alpha_i} = \sum_{\substack{\beta \in \mathbb{Z}_+^n \\ |\beta| = \alpha_i}} \frac{\alpha_i!}{\beta!} x^\beta \quad 1 \leq i \leq n,$$

so that each factor on the right side of 2.1 is a linear combination of homogeneous polynomials of degree α_i , $1 \leq i \leq n$. By the definition of the product of polynomials we will have that 2.1 is a linear combination of monomials of degree $\alpha_1 + \cdots + \alpha_n = m$ and therefore the monomials are invariant under the action of the group $O(n)$. ■

Proposition 2.1.8. *The space $\mathcal{H}_m(\mathbb{R}^n)$ is an invariant subspace of the quasi-regular representation of $O(n)$ on $\mathbb{C}^{\mathbb{R}^n}$.*

Proof. Let $g = [g_{ij}]_{ij} \in O(n)$. We know that by the chain rule

$$\frac{\partial}{\partial x_j}(f \circ L_g) = \sum_{i=1}^n \partial_i f \cdot \frac{\partial}{\partial x_j} y_i = \sum_{i=1}^n \partial_i f \cdot g_{ij},$$

where $y_i = \sum_{k=1}^n g_{ik}x_k$. Differentiating again we have that

$$\frac{\partial^2}{\partial x_j^2}(f \circ L_g) = \frac{\partial}{\partial x_j} \left(\sum_{i=1}^n \partial_i f g_{ij} \right) = \sum_{i=1}^n g_{ij} \frac{\partial}{\partial x_j} (\partial_i f) = \sum_{i=1}^n g_{ij} \sum_{k=1}^n \partial_k \partial_i f g_{kj} = \sum_{i,k=1}^n g_{ij} g_{kj} \partial_k \partial_i f.$$

Now calculating the Laplacian we have that

$$\Delta(f \circ L_g) = \sum_{j=1}^n \frac{\partial^2}{\partial x_j^2}(f \circ L_g) = \sum_{j=1}^n \left(\sum_{i,k=1}^n g_{ij} g_{kj} \partial_k \partial_i f \right) = \sum_{j=1}^n \left(\sum_{i,k=1}^n g_{ij} g_{jk}^T \partial_k \partial_i f \right).$$

From the fact that g is orthogonal we have that $\sum_{j=1}^n g_{ij} g_{jk}^T = \delta_{ik}$, since this sum is the (i, k) entry of the product $gg^T = Id_n$. From the previous comment we have that

$$\Delta(f \circ L_g) = \sum_{i,k=1}^n \delta_{ik} \partial_k \partial_i f = \sum_{i=1}^n \partial_i^2 f = 0. \blacksquare$$

Analogous results hold for the spaces $\mathcal{H}_m(S^{n-1})$ and $\mathcal{P}_m(S^{n-1})$.

If we denote by ρ_m and π_m the quasi-regular representations of the group $SO(n)$ on $\mathcal{P}_m(\mathbb{R}^n)$ and $\mathcal{P}_m(S^{n-1})$, respectively, we have that the restriction operator is an intertwining operator.

Proposition 2.1.9. *The restriction operator, R , is an intertwining operator.*

Proof. Let us see that for each $g \in G$, the following diagram commutes

$$\begin{array}{ccc} \mathcal{P}_m(\mathbb{R}^n) & \xrightarrow{\rho_m(g)} & \mathcal{P}_m(\mathbb{R}^n) \\ \downarrow R & & \downarrow R \\ \mathcal{P}_m(S^{n-1}) & \xrightarrow{\pi_m(g)} & \mathcal{P}_m(S^{n-1}), \end{array}$$

since for each $p \in \mathcal{P}_m(\mathbb{R}^n)$, we have that

$$p|_{S^{n-1}} \circ L_{g^{-1}} = (p \circ L_{g^{-1}})|_{S^{n-1}}. \blacksquare$$

In particular, this shows that these representations are equivalent. Once the equivalence of these two representations is established, it is sufficient to prove the irreducibility of one of them to establish the irreducibility of both. We will do this using Lemma 4.0.1, but for this we need to define suitable inner products on our polynomial spaces. There are two possible inner products for the space of spherical harmonics, both with their advantages. We will start by defining the inner product of the sphere.

Definition 2.1.10 (Inner product of the sphere). Let $d\sigma$ be the probability measure on S^{n-1} invariant with respect to $\text{SO}(n)$, i.e.,

$$d\sigma(S^{n-1}) = 1.$$

Such a measure exists by Proposition 1.4.4. With respect to this measure, we define the inner product of the sphere as

$$\langle f, g \rangle_{S^{n-1}} = \int_{S^{n-1}} f \bar{g} d\sigma.$$

Note that this is the inner product of $L^2(S^{n-1}, d\sigma)$. We know that by the continuity of the polynomials and by having compact support, we have that

$$\mathcal{H}_m(S^{n-1}) \subset L^2(S^{n-1}, d\sigma),$$

moreover, being finite-dimensional, it is an RKHS.

Proposition 2.1.11. The quasi-regular representation of $\text{SO}(n)$ on $\mathcal{H}_m(S^{n-1})$ with the inner product of the sphere is unitary.

Proof. Since $\mathcal{H}_m(S^{n-1})$ is a finite-dimensional Hilbert space, it suffices to prove that $\pi(g)$ preserves the inner product $\forall g \in G$. Let $p, q \in \mathcal{H}_m(S^{n-1})$, let us see that

$$\begin{aligned} \langle \pi(g)p, \pi(g)q \rangle_{S^{n-1}} &= \int_{S^{n-1}} \pi(g)p(x) \overline{\pi(g)q(x)} d\sigma(x) \\ &= \int_{S^{n-1}} \pi(g)p \bar{q}(x) d\sigma(x) \\ &= \int_{S^{n-1}} p(x) \overline{q(x)} d\sigma(x) = \langle p, q \rangle_{S^{n-1}}, \end{aligned}$$

which shows that $\pi(g)$ is an isometry and, therefore, a unitary operator. $\blacksquare$

Since $\mathcal{H}_m(S^{n-1})$ is a finite-dimensional vector subspace of $L^2(S^{n-1}, d\sigma)$, the following result is evident.

Proposition 2.1.12. The space $\mathcal{H}_m(S^{n-1})$ with the inner product of the sphere is an RKHS.

It is also possible to define an inner product known as the **Fischer product**. With this inner product, the space $\mathcal{H}_m(\mathbb{R}^n)$ becomes an RKHS.

Definition 2.1.13. Given $p, q \in \mathcal{P}_m(\mathbb{R}^n)$ with

$$p = \sum_{|\alpha|=m} a_\alpha x^\alpha,$$

we define the **Fischer-Bombieri** product as

$$\langle p, q \rangle_F = p(\partial)(\bar{q}),$$

where $p(\partial)$ or ∂_p is the notation for the differential operator

$$\sum_{|\alpha|=m} a_\alpha \partial^\alpha.$$

This remains an inner product when restricted to homogeneous harmonics. Furthermore, it has the advantage of being easier to compute than the product of the sphere.

Lema 2.1.14. Let $p, q \in \mathcal{P}_m(\mathbb{R}^n)$ with

$$p = \sum_{|\alpha|=m} a_\alpha x^\alpha \quad \text{and} \quad q = \sum_{|\beta|=m} b_\beta x^\beta,$$

then

$$\langle p, q \rangle_F = \sum_{|\alpha|=m} a_\alpha \bar{b}_\alpha \alpha! \quad (2.2)$$

Proof. First let us note that

$$\partial^\alpha(x^\beta) = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}}(x^\beta) = \begin{cases} 0 & \text{if } \alpha \neq \beta \\ \alpha! & \text{if } \alpha = \beta. \end{cases}$$

By the definition of the Fischer product and the linearity of the differential operators, we have that

$$\langle p, q \rangle_F = \sum_{|\alpha|=m} a_\alpha \partial^\alpha \left(\sum_{|\beta|=m} \bar{b}_\beta x^\beta \right) = \sum_{|\alpha|=m} a_\alpha \sum_{|\beta|=m} \bar{b}_\beta \partial^\alpha(x^\beta) = \sum_{|\alpha|=m} a_\alpha \bar{b}_\alpha \alpha! \blacksquare$$

Proposition 2.1.15. The Fischer-Bombieri product is an inner product on the space $\mathcal{P}_m(\mathbb{R}^n)$ and on the space $\mathcal{H}_m(\mathbb{R}^n)$.

Proof. This follows directly from (2.2) and the positivity of the coefficients of the sum. $\blacksquare$

In particular, we are interested in the restriction of this inner product to the space $\mathcal{H}_m(\mathbb{R}^n)$.

Theorem 2.1.16. With the Fischer product, $\mathcal{H}_m(\mathbb{R}^n)$ is an RKHS.

This follows because $\mathcal{H}_m(\mathbb{R}^n)$ is a finite-dimensional vector space. We will show that with this product the quasi-regular representation becomes unitary.

Theorem 2.1.17. The Fischer product on $\mathcal{H}_m(\mathbb{R}^n)$ is $\text{SO}(n)$ -invariant.

Proof. For $f \in \mathbb{C}^{\mathbb{R}^n}$ and $g \in \text{SO}(n)$, we fix the notation

$$(g \cdot f)(x) = (\rho(g) \cdot f)(x) = f(g^{-1} \cdot x).$$

Given a polynomial

$$p(x) = \sum_{|\alpha|} a_\alpha x^\alpha$$

we establish the notation

$$p(\partial) = \partial_p = \sum_{|\alpha|} a_\alpha \partial^\alpha.$$

We express each matrix $g \in \text{SO}(n)$ in terms of its entries and rows, the same applies to the matrix $A = g^{-1}$, then

$$g = (g_{jk})_{j,k=\overline{1,n}} = \begin{pmatrix} g_1 \\ \vdots \\ g_n \end{pmatrix}$$

with g_i the i -th row of the matrix g . Analogously

$$g^{-1} = A = (a_{jk}) = \begin{pmatrix} A_1 \\ \vdots \\ A_n \end{pmatrix}$$

with A_i the i -th row of the matrix A .

For $g \in \text{SO}(n)$ and smooth $f \in \mathbb{C}^{\mathbb{R}^n}$, it holds that

$$\partial_{x_k}(g \cdot f) = g \cdot \partial_{\langle g_k, x \rangle} f = g \cdot \partial_{g^{-1} \cdot x_k} f,$$

where

$$\langle g_k, x \rangle = g_{k,1}x_1 + \cdots + g_{k,n}x_n.$$

Then

$$\partial^\alpha(g \cdot f) = g \cdot (\partial_{\langle g_1, x \rangle}^{\alpha_1} \cdots \partial_{\langle g_n, x \rangle}^{\alpha_n} f),$$

where $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$. Indeed, we calculate

$$\begin{aligned} \partial_{x_k}((g \cdot f)(x)) &= \frac{\partial}{\partial x_k}(f(g^{-1} \cdot x)), \quad \text{with} \quad g^{-1} \cdot x = \begin{pmatrix} A_1 x \\ \vdots \\ A_n x \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \\ &= \sum_{j=1}^n \frac{\partial f}{\partial y_j}(g^{-1} \cdot x) \frac{\partial y_j}{\partial x_k}, \quad y_j = a_{j,1}x_1 + \cdots + a_{j,n}x_n \\ &= \sum_{j=1}^n \frac{\partial f}{\partial y_j}(g^{-1} \cdot x) a_{jk}, \quad a_{jk} = (g^{-1})_{jk} = (g^T)_{j,k} = g_{k,j} \\ &= \sum_{j=1}^n g_{kj} \frac{\partial f}{\partial x_j}(g^{-1} \cdot x), \end{aligned}$$

this is rewritten as

$$\partial_{x_k}(g \cdot f)(x) = (g \cdot \partial_{\langle g_k, x \rangle} f)(x)$$

or else

$$\partial_{x_k}(g \cdot f) = g \cdot \partial_{\langle g_k, x \rangle} f.$$

Applying the differential operator ∂_{x_k} again to the previous result we have that

$$\partial_{x_k}^2(g \cdot f) = \partial_{x_k}[\partial_{x_k}(g \cdot f)] = \partial_{x_k}[g \cdot \partial_{\langle g_k, x \rangle} f] = g \cdot \partial_{x_k}[\partial_{\langle g_k, x \rangle} f] = g \cdot \partial_{\langle g_k, x \rangle}^2 f.$$

Continuing this process we have that for $l \in \mathbb{N}$,

$$\partial_{x_k}^l(g \cdot f) = g \cdot \partial_{\langle g_k, x \rangle}^l f,$$

therefore,

$$\partial^\alpha(g \cdot f) = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n}(g \cdot f) = g \cdot (\partial_{\langle g_1, x \rangle}^{\alpha_1} \cdots \partial_{\langle g_n, x \rangle}^{\alpha_n} f).$$

Note that the expression on the right side is precisely

$$\partial_{g^{-1} \cdot x^\alpha} f.$$

Now, by the linearity of the representation and because it is an inner product, it suffices to prove the invariance only for monomials.

$$\langle g \cdot x^\alpha, g \cdot x^\beta \rangle = \partial^\alpha(g^{-1} \cdot (g \cdot x^\beta)) = g^{-1} \cdot \partial_{g \cdot x^\alpha}(g \cdot x^\beta) = g \cdot \langle g \cdot x^\alpha, g \cdot x^\beta \rangle = \langle x^\alpha, x^\beta \rangle. \blacksquare$$

Now that we have two inner products, each of which makes the spaces $\mathcal{H}_m(\mathbb{R}^n)$ and $\mathcal{H}_m(S^{n-1})$

RKHSs, we can verify the irreducibility of the representations using Theorem 4.0.1 in either of these two Hilbert spaces. Before concluding, let us see that with the Fischer inner product it is easy to show that the space $\mathcal{P}_m(\mathbb{R}^n)$ is not irreducible.

Proposition 2.1.18. *For $n \geq 2$, the subspace*

$$|\cdot|^2 \mathcal{P}_{m-2}(\mathbb{R}^n)$$

is a proper $O(n)$ -invariant subspace of $\mathcal{P}_m(\mathbb{R}^n)$.

Proof. Let $h_{m-2} \in \mathcal{P}_{m-2}(\mathbb{R}^n)$, $g \in O(n)$ and $x \in \mathbb{R}^n$ be arbitrary. Since g is an orthogonal matrix and $\mathcal{P}_{m-2}(\mathbb{R}^n)$ is $O(n)$ -invariant, it holds that

$$\begin{aligned} \rho_m(g)(|\cdot|^2 h_{m-2})(x) &= |g^{-1}x|^2 h_{m-2}(g^{-1}x) \\ &= |x|^2 \rho_{m-2}(g)(h_{m-2})(x) = |x|^2 s_{m-2}(x), \end{aligned}$$

where $s_{m-2} \in \mathcal{P}_{m-2}(\mathbb{R}^n)$. Therefore, we conclude that $\rho_m(g)(|\cdot|^2 h_{m-2}) \in |\cdot|^2 \mathcal{P}_{m-2}(\mathbb{R}^n)$. ■

We also have the following orthogonal direct sum decomposition:

Proposition 2.1.19. $\mathcal{P}_m(\mathbb{R}^n) = \mathcal{H}_m(\mathbb{R}^n) \oplus |\cdot|^2 \mathcal{P}_{m-2}(\mathbb{R}^n)$.

Proof. With the Fischer inner product for $\mathcal{P}_m(\mathbb{R}^n)$ we show that

$$\mathcal{H}_m(\mathbb{R}^n) \perp |\cdot|^2 \mathcal{P}_{m-2}(\mathbb{R}^n).$$

First let us prove the containment $\mathcal{H}_m(\mathbb{R}^n) \subset (|\cdot|^2 \mathcal{P}_{m-2}(\mathbb{R}^n))^\perp$. Let $h \in \mathcal{H}_m(\mathbb{R}^n)$ be arbitrary. For each $p_{m-2} \in \mathcal{P}_{m-2}(\mathbb{R}^n)$ we have that

$$\langle |\cdot|^2 p_{m-2}, h \rangle_F = p_{m-2}(\partial)(|\partial|^2 \bar{h}) = p_{m-2}(\partial)(\Delta \bar{h}) = 0,$$

then $h \in (|\cdot|^2 \mathcal{P}_{m-2}(\mathbb{R}^n))^\perp$. Now let us prove the other containment. Let $h \in (|\cdot|^2 \mathcal{P}_{m-2}(\mathbb{R}^n))^\perp$ be arbitrary. For each $p_{m-2} \in \mathcal{P}_{m-2}(\mathbb{R}^n)$ we have that

$$0 = \langle |\cdot|^2 p_{m-2}, h \rangle_F = p_{m-2}(\partial)(\Delta \bar{h}) = \langle p_{m-2}, \Delta h \rangle_F,$$

where the last inner product is the Fischer inner product in the space $\mathcal{P}_{m-2}(\mathbb{R}^n)$. Since $\Delta h \in \mathcal{P}_{m-2}(\mathbb{R}^n)$ and p_{m-2} is arbitrary, we have that $\Delta h = 0$. ■

Example 12. The Hilbert space $L^2(\cdot, d\mu)$ is not an RKHS, primarily because it is not a Hilbert space of functions; its elements are equivalence classes of functions; furthermore, assuming there was a way to choose a representative of each class such that the evaluation functional is well-defined, it is possible to show that this would not be continuous.

2.2 Main results

The main result that allows us to connect representation theory and the theory of **RKHS** lies in the reproducing properties of the latter, as well as in the fact that these spaces are uniquely determined by their kernel.

Proposition 2.2.1. *Let $\mathcal{H}$ be an RKHS and $K(\cdot, \cdot)$ the reproducing kernel. For $x, y \in X$, the kernel satisfies*

1. $K_y := K(\cdot, y) \in \mathcal{H}$,
2. $\overline{K(x, y)} = K(y, x) = \langle K_x, K_y \rangle$,
3. $K(y, y) = \|K_y\|^2 = \|K(\cdot, y)\|^2$.

Proof.

1. Given $y \in X$, we have that $K(\cdot, y) = K_y \in \mathcal{H}$ by the Riesz representation theorem.
2. $K(x, y) = K_y(x) = F_x(K_y) = \langle K_y, K_x \rangle = \overline{\langle K_x, K_y \rangle} = \overline{K_x(y)} = \overline{K(y, x)}$.
3. $\|K_y\|^2 = \langle K_y, K_y \rangle = K_y(y) = K(y, y)$.

From the kernels we can obtain a total set for the Hilbert space.

Theorem 2.2.2. *Let $\mathcal{H}$ be an RKHS over the set X and $K : X \times X \rightarrow \mathbb{C}$ the reproducing kernel, then the collection*

$$\{K_y(\cdot) = K(\cdot, y) : y \in X\}$$

generates a dense subspace in $\mathcal{H}$.

Proof. Let $H_0 = \text{span}\{K_y(\cdot)\}_{y \in X}$. We claim that $\overline{H_0} = \mathcal{H}$. Let us recall that

$$(\overline{H_0})^\perp = H_0^{\perp\perp} = \overline{H_0^\perp} = H_0^\perp,$$

so it is sufficient to prove that $H_0^\perp = \{0\}$. Let $f \in H_0^\perp$, it holds that para todo $y \in X$

$$f(y) = \langle f, K_y \rangle = 0 \quad \text{para todo } y \in X$$

and therefore $f \equiv 0$. ■

One of the properties we have in an RKHS is that convergence in the Hilbert space implies pointwise convergence.

Lema 2.2.3. *Let $\mathcal{H}$ be an RKHS over X . If $\{f_n\}_{n=1}^\infty$ converges to f in $\mathcal{H}$, then $\{f_n\}_{n=1}^\infty$ converges pointwise to f , that is, for each $x \in X$ we have that*

$$\{f_n(x)\}_{n=1}^\infty \rightarrow f(x).$$

Proof. Let $x \in X$, note that

$$|f(x) - f_n(x)| = |F_x(f - f_n)| = |\langle f - f_n, K_x \rangle| \leq \|f - f_n\| \|K_x\|$$

by the definition of the kernel and the Cauchy-Schwarz inequality. Given $\epsilon_0 = \frac{\epsilon}{1 + \|K_x\|}$, there exists $N \in \mathbb{N}_0$ such that $\|f_n - f\| < \epsilon_0$ for all $n > N$. Therefore,

$$|f(x) - f_n(x)| \leq \frac{\epsilon}{1 + \|K_x\|} \cdot \|K_x\| < \epsilon.$$

The following result is one of the remarkable features of the theory of RKHS.

Theorem 2.2.4. *Let $\mathcal{H}$ be an RKHS and $\{e_i\}_{i \in I}$ an orthogonal basis for $\mathcal{H}$. Then*

$$K(x, y) = \sum_{i \in I} e_i(x) \overline{e_i(y)},$$

where the convergence is pointwise convergence. It also holds that

$$K_y = \sum_{i \in I} e_i \overline{e_i(y)},$$

where the convergence occurs in the metric of $\mathcal{H}$.

Proof. Since it is an orthonormal basis we have that

$$K(\cdot, y) = K_y = \sum_{j \in J} \langle K_y, e_j \rangle e_j = \sum_{j \in J} \overline{e_j(y)} e_j.$$

Since convergence in an RKHS implies pointwise convergence we have that

$$K(x, y) = K_y = \sum_{j \in J} \langle K_y, e_j \rangle e_j(x) = \sum_{j \in J} \overline{e_j(y)} e_j(x). \blacksquare$$

Theorem 2.2.5. *Let $\mathcal{H}$ be an RKHS and $\mathcal{H}_0 \subseteq \mathcal{H}$ a closed subspace. It holds that $\mathcal{H}_0$ is an RKHS and that the kernel of $\mathcal{H}_0$, denoted by Φ , is*

$$\Phi(x, y) = \langle P_0 K_y, K_x \rangle \quad \text{para todo } x, y \in X,$$

where $P_0 : \mathcal{H} \rightarrow \mathcal{H}$ is the orthogonal projection from $\mathcal{H}$ onto $\mathcal{H}_0$.

Proof. For each $x \in X$, the evaluation operator

$$\hat{F}_x : \mathcal{H}_0 \rightarrow \mathbb{C}$$

is the restriction to $\mathcal{H}_0$ of the evaluation operator F_x , which by hypothesis is continuous. Since $\mathcal{H}_0$ is closed, the operator $\hat{F}_x$ is continuous $\forall x \in X$; therefore, $\mathcal{H}_0$ is an RKHS.

Now let Φ be the reproducing kernel of $\mathcal{H}_0$. Note that

$$\hat{F}_x(f) = \langle f, \Phi_x \rangle \quad \text{para todo } f \in \mathcal{H}_0 \text{ para todo } x \in X.$$

On the other hand, for each $f \in \mathcal{H}_0$ and $x \in X$ we have that

$$\begin{aligned} \hat{F}_x(f) &= F_x(f) = \langle f, K_x \rangle \\ &= \langle P_0 f, K_x \rangle = \langle f, P_0^* K_x \rangle = \langle f, P_0 K_x \rangle, \end{aligned}$$

since P_0 is Hermitian for being an orthogonal projection. By the uniqueness of the Riesz-Fréchet representation theorem we have that

$$\Phi_x = P_0 K_x \quad \text{para todo } x \in X,$$

from which we conclude that

$$\Phi(x, y) = \langle P_0 K_y, K_x \rangle. \blacksquare$$

This last result is fundamental for the proof of the main result of the next section.

Theorem 2.2.6. *Let $\mathcal{H}_1, \mathcal{H}_2$ be reproducing kernel Hilbert spaces on X , with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ respectively, let $K_1, K_2 : X \times X \rightarrow \mathbb{C}$ be the respective reproducing kernels. If*

$$K_1(x, y) = K_2(x, y) \quad \text{para todo } x, y \in X,$$

then $\mathcal{H}_1 = \mathcal{H}_2$ and $\|\cdot\|_1 = \|\cdot\|_2$.

Proof. We define $K := K_1 = K_2$. Let us consider the spaces

$$W_i = \text{span} \{K_{y_i}\}_{y \in X} \quad i = 1, 2.$$

We have that $W_1 = W_2$. Furthermore $\mathcal{H}_1 = \overline{W_1}$ and $\mathcal{H}_2 = \overline{W_2}$ with the respective topologies of the spaces $\mathcal{H}_1$ and $\mathcal{H}_2$. We will prove that both spaces have the same norm and therefore both closures are identical; we will conclude that $\mathcal{H}_1 = \mathcal{H}_2$.

Note that any $f \in W_1$ is expressed as

$$f = \sum_{j=1}^m a_j K_{y_j},$$

so that,

$$\|f\|_1^2 = \langle f, f \rangle_1 = \left\langle \sum_{j=1}^m a_j K_{y_j}, \sum_{k=1}^m a_k K_{y_k} \right\rangle_1 = \sum_{j,k=1}^m a_j \overline{a_k} K(y_k, y_j).$$

Analogously, we have

$$\|f\|_2^2 = \sum_{j,k=1}^m a_j \overline{a_k} K(y_k, y_j).$$

So that $\|f\|_1 = \|f\|_2$ para todo $f \in W_1 = W_2$.

Now, for each $f \in \mathcal{H}_1$ there exists a sequence of elements in W_1 , $\{f_n\}_{n \in \mathbb{N}}$, such that

$$\lim_{n \rightarrow \infty} \|f - f_n\|_1 = 0.$$

In particular $\{f_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $\|\cdot\|_1$ and, therefore, also in $\|\cdot\|_2$, so there exists $g \in \mathcal{H}_2$ such that

$$\lim_{n \rightarrow \infty} \|g - f_n\|_2 = 0.$$

Since convergence in an RKHS implies pointwise convergence, we have that

$$g(x) = \lim_{n \rightarrow \infty} f_n(x) = f(x) \quad (\forall x \in X),$$

therefore f and g are identical.

Finally, the continuity of the norm guarantees us that

$$\|f\|_1 = \|\lim_{n \rightarrow \infty} f_n\|_1 = \lim_{n \rightarrow \infty} \|f_n\|_1 = \lim_{n \rightarrow \infty} \|f_n\|_2 = \|f\|_2. \blacksquare$$

3 Basic Representation Theory

Objectives:

1. Formalize the action of a group on a vector space, be it a Hilbert or Banach space
2. Define what a unitary representation is.
3. Give the example of the quasi-regular representation
4. Perhaps give a motivation for the study of representations

3.1 Preliminaries and unitary representations

Definition 3.1.1. Let G be a group and V a non-zero vector space over a field $\mathbb{F}$. A representation of G on V is a group homomorphism

$$\pi : G \rightarrow \text{Aut}(V),$$

where $\text{Aut}(V) = \{T : V \rightarrow V : T \text{ is linear and invertible}\}$. If V is a finite-dimensional vector space, we will say that **the representation is finite-dimensional**, otherwise we will say that **the representation is infinite-dimensional**.

In the present work we focus on the so-called **unitary representations**; for this we will always consider G as a topological group and $\mathcal{H}$ a Hilbert space. A **unitary representation** of G on $\mathcal{H}$ is a group homomorphism

$$\pi : G \rightarrow \mathcal{U}(\mathcal{H}).$$

If, in addition, the homomorphism is a continuous function when $\mathcal{U}(\mathcal{H})$ has the strong operator topology, we will say that the representation is **strongly unitary**.

We must emphasize that this is only one type of representation; this type of representations is used to be able to study topological groups and their actions with the theory of *functional analysis*.

Example 13 (Standard representation). Let $G = \text{SO}(n)$. The standard representation is the representation of G on $\mathbb{R}^n$, where $\pi(g)$ is given by left multiplication by the matrix g , that is,

$$\pi(g)(x) = gx.$$

Example 14. Consider the group $\mathbb{R}$ under addition, and the Hilbert space $\mathbb{R}^2$ with the usual inner product. We define the unitary representation $\pi : \mathbb{R} \rightarrow \mathcal{U}(\mathbb{R}^2)$ by means of

$$\pi(x) = \begin{bmatrix} \cos(x) & -\sin(x) \\ \sin(x) & \cos(x) \end{bmatrix}.$$

Example 15 (Regular representation). Let G be a compact group and ρ the left Haar measure associated with the group. **The left regular representation** on the Hilbert space $L^2(G, \rho)$ is defined by translations, that is,

$$[\pi_L(g)f](h) = L_g f(h) = f(g^{-1}h) \quad (\forall g, h \in G)(\forall f \in L^2(G, \rho)).$$

This is well defined because given $f \in L^2(G, \rho)$ we have that

$$\int_G |\pi_L(g)f(h)|^2 d\rho(h) = \int_G |L_g f(h)|^2 d\rho(h) = \int_G |f(g^{-1}h)|^2 d\rho(h) = \int_G |f(h)|^2 d\rho(h) < \infty$$

by the invariance of the Haar measure.

Fixing $g \in G$ we have that for each $\alpha \in \mathbb{R}$ and $f_1, f_2 \in L^2(G, \rho)$ it holds that

$$L_g(\alpha f_1 + f_2)(h) = \alpha f_1(g^{-1}h) + f_2(g^{-1}h) = \alpha L_g f_1(h) + L_g f_2(h) \quad \text{para todo } h \in G,$$

so $\pi_L(g)$ is a linear transformation for each $g \in G$. Furthermore it has an inverse given by $\pi_L(g^{-1})$ which proves that in fact π_L is a representation. Moreover, it is a unitary representation.

The following example plays a core role in representation theory, for pedagogical reasons I will only focus on the case of G a Lie group although this representation can be fulfilled for the case of G locally compact.

Example 16 (Quasi-regular representation). Let G be a compact Lie group and H a closed subgroup. We know from Theorem 1.4.4 that the quotient G/H has an invariant Haar measure μ , we define an action of the group G on the Hilbert space $L^2(G/H, \mu)$ as follows

$$\begin{aligned}\pi : G &\rightarrow \mathcal{U}(L^2(G/H, \mu)) \\ g &\mapsto L_g\end{aligned}$$

where $L_g f(xH) = f(gxH)$.

A simpler (but less elementary) way of saying this is to express the quasi-regular representation as an induced representation. Concretely, this representation is $\text{ind}_H^G(\rho)$ where ρ is the trivial representation of the group H on $\mathbb{C}$ (*A Course in Abstract Harmonic Analysis* [Fol94], page 165).

We can give a concrete example with the homogeneous spaces we have already covered.

Example 17. The groups $O(n+1)$ and $SO(n+1)$ act on S^n transitively by means of left multiplication. So using the invariant measure we can define the quasi-regular representation of the groups $O(n)$ and $SO(n)$ on S^n

3.2 Irreducible representations.

Similar to the study of operators we seek to understand a representation through its behavior on smaller spaces, this with the purpose of facilitating the study of the representation. This gives rise to the concept of invariant spaces of the representation.

Definition 3.2.1. Let $\mathcal{M}$ be a closed subspace of the Hilbert space $\mathcal{H}_\pi$. It will be said that $\mathcal{M}$ is invariant for a representation $\pi : G \rightarrow \mathcal{U}(\mathcal{H}_\pi)$ or G -invariant if

$$\pi(x)\mathcal{M} \subset \mathcal{M} \quad \text{para todo } x \in G,$$

that is, if it is an invariant subspace of the operator $\pi(x)$ para todo $x \in G$.

Let $\mathcal{M} \neq \{0\}$ be a closed and invariant subspace for the representation π , the restriction of the representation π to $\mathcal{M}$ denoted by $\pi^\mathcal{M}$ is the homomorphism $\pi^\mathcal{M} : G \rightarrow \mathcal{U}(\mathcal{M})$ defined by

$$\pi^\mathcal{M}(g) = \pi(g)|_\mathcal{M}.$$

It is evident that this restriction is again a representation.

Regarding the type of invariant subspaces on which we will focus, they are the **irreducible** subspaces of the representation.

Definition 3.2.2. Let π be a representation of a group G on a Hilbert space $\mathcal{H}$, the representation is said to be **irreducible** if it has no closed and invariant subspaces other than $\{0\}$ and $\mathcal{H}$.

““latex

4 Representations in G -Hilbert spaces

Objectives:

1. Define G -Hilbert spaces
2. Give the strong theorem that connects the theory of $RKHS$ and the theory of unitary representations.
3. Define what a Gelfand pair is
4. Define spherical functions
5. Zonal spherical functions and some classical polynomials
6. Explain the theorems related to this concept
7. Express some reproducing kernels in terms of spherical functions (zonal spherical).

Consider $\mathcal{H}$ a reproducing kernel Hilbert space of functions on X , and consider a group G acting on the space X . There is a canonical action of the group G on the space $\mathbb{C}^X$ defined by the following formula

$$[\pi(g)f](x) = f(g^{-1} \cdot x) \quad \text{para todo } x \in X \text{ and para todo } g \in G$$

We define a G -Hilbert space of functions on X as a Hilbert space of functions on X , such that

$$\pi(g)\psi \in \mathcal{H} \quad \text{and } \|\pi(g)\psi\| = \|\psi\|$$

It is evident that with the previous action we can restrict the

Theorem 4.0.1 (Proposition 2, [BdG03]). *With the notation previously used, we have that the following statements hold*

1. $\pi(g)K_x = K_{gx}$,
2. $K(gx, gy) = K(x, y)$ para todo $g \in G$ and para todo $x, y \in X$
3. Suppose that G acts transitively on X (i.e., X is a homogeneous space), and we take $w \in X$ arbitrary but fixed. Consider its isotropy group $F := G^w$, then it holds that:
 - If $\mathcal{H} \neq \{0\}$, then $\mathcal{H}^F \neq \emptyset$.
 - If $\dim \mathcal{H}^F = 1$, then the representation π is irreducible.

Definition 4.0.2. An irreducible representation $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ will be called of class one with respect to the subgroup K if it holds that

$$\dim_{\mathbb{C}} \mathcal{H}^K = 1$$

There is a partial converse of this result that remains valid for the case of Hilbert spaces. However, we must define a couple of notions before we can state our proposition.

4.1 Bi-invariant functions

Definition 4.1.1. Let G be a group and K a subgroup. A function f is said to be K -invariant if

$$f(k_1 x k_2) = f(x)$$

for every $k_1, k_2 \in K$ and $x \in G$.

We denote by $C_c(G//K)$ the set of compactly supported bi-invariant functions. This set together with the convolution operation is a function algebra (I am not sure if it is a $*$ -algebra); recall that the convolution for groups is defined as

$$f * \psi(x) = \int_G f(xy^{-1})\psi(y)dy.$$

It is very important to note that the algebra $C_c(G//K)$ is a subalgebra of $L^1(G)$. We know that the algebra $L^1(G)$ is abelian if and only if G is abelian. The abelian subalgebras of the algebra $L^1(G)$ are of importance because...

Definition 4.1.2. Let G be a locally compact Hausdorff group and K a compact subgroup. We say that (G, K) is a Gelfand pair if the algebra $C_c(G//K)$ is abelian.

Although equivalences of this definition can be found, for pedagogical reasons we will stick with this one for the remainder of the course. It is also worth mentioning that definitions of Gelfand pairs for other types of groups can be found in the literature.

Theorem 4.1.3. Let $\pi : G \rightarrow \text{GL}(\mathcal{H})$ be a representation that is star-closed in G and in K . Suppose that $\mathcal{H}^k \neq 0$ and that $\mathcal{H}$ is equal to the closure of $\pi(G)\mathcal{H}^k$. Then $\mathcal{H}^k$ is $C_c(G//K)$ -irreducible if and only if $\mathcal{H}$ is $C_c(G)$ -irreducible.

The following result is an almost immediate consequence of the previous theorem; its importance lies in showing the strong relationship between the algebra $C_c(G//K)$ and the theory of unitary representations.

Theorem 4.1.4. Let $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ be an irreducible unitary representation. If the algebra $C_c(G//K)$ is abelian, then $\dim_{\mathbb{C}} \mathcal{H}^K \leq 1$

Now that we have this theorem, we can give the conditions to find the converse of Theorem 4.0.1.

Theorem 4.1.5 (Theorem for Doctor Francisco). Let G be an LC group and K a compact subgroup. Assume that (G, K) is a Gelfand pair. Assume that the space $\mathcal{H}^K$ is cyclic. Then the converse of Theorem 4.0.1 holds.

4.2 Spherical functions and their connection to unitary representations

This section seeks to fulfill the following objectives

1. To be able to explain in which cases a reproducing kernel can be expressed in terms of spherical functions. (for Gera)
2. To be able to explain how special functions derive from representations (for the students)
3. To be able to give a final demonstration that this topic also connects with unitary representations and with matrix entries

Following the structure proposed by Lang in $SL_2(R)$, we are going to define a spherical function as that function on G that has the spherical property.

Definition 4.2.1 (The spherical property). Let G be a unimodular group and $K \subset G$ a compact subgroup. We say that a function f is K -spherical, or spherical, if it satisfies the following properties:

1. f is bi-invariant and continuous.
2. f is a left eigenfunction of the algebra $C_c(G//K)$ i.e.,

$$f * \psi = \lambda(f, \psi)f \quad \forall \psi \in C_c(G//K)$$

3. $f(e) = 1$

Let $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation of the group G on the Hilbert space $\mathcal{H}$. Let $u \in \mathcal{H}^K$ be unitary. We will consider the following matrix entry

$$f(x) = \pi_{u,u}(x) = \langle \pi(x)u, u \rangle$$

this function is bi-invariant and also satisfies $f(e) = 1$.

Theorem 4.2.2 ($SL_2(R)$ -Lang, Theorem 8). *Let $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation. Suppose there exists a vector $u \in \mathcal{H}^K$ that topologically generates the space $\mathcal{H}$ via π . Then we have that*

$$\dim \mathcal{H}^K = 1 \quad \Leftrightarrow \quad \text{the function } f(x) = \langle \pi(x)u, u \rangle \text{ is spherical}$$

4.3 Reproducing kernels and spherical functions

We are going to cover section 5 of the paper “Irreducibility of unitary group representations and reproducing kernels Hilbert spaces” by Bekka, de la Harpe, and Grigorchuk. This section addresses a case in which we can express the reproducing kernel of a space in terms of spherical functions. In that work, they focus only on the case where G is a compact group and K is a closed subgroup. The homogeneous space $X = G/K$ is considered together with a G -invariant and positive measure μ on X , denoting by π the natural unitary representation of the group G on the space $L^2(X, \mu)$.

Suppose that the representation is of class one with respect to the subgroup K . A vector $\eta \in \mathcal{H}^K$ is chosen. The **zonal spherical** function with respect to η is defined as

$$f(g) = \langle \pi(g)\eta, \eta \rangle$$

let us see that this function is K -bi-invariant. Let $\Phi : X \times X \rightarrow \mathbb{C}$ be the reproducing kernel of the G -Hilbert space $\mathcal{H}$. Let us denote by w the canonical base point (the class of the group K) and we define the function

$$\begin{aligned} \phi_w : X &\rightarrow \mathbb{C} \\ x &\mapsto \Phi(x, w) \end{aligned}$$

both functions ϕ_w, η are in $\mathcal{H}^K$, so there exists a non-zero complex c such that

$$\phi_w = c\eta$$

Theorem 4.3.1. *With the previous notation we have that*

$$\Phi(gw, g'w) = \frac{\dim_{\mathbb{C}} \mathcal{H}}{\mu(X)} f(g^{-1}g') \quad \text{and} \quad \eta(gw) = \sqrt{\frac{\dim_{\mathbb{C}} \mathcal{H}}{\mu(X)}} f(g^{-1})$$

Let us see that this theorem allows us to cover the case of $S^n = \text{SO}(n)\text{SO}(n-1)$

Example 18. If $G = \text{SO}(3)$ and $K = \text{SO}(2)$, we have that the function f is a Legendre polynomial.

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